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Slippers2Sat-2 (S2S-2): ADCS

S2S-2 Team

Kathmandu, Nepal



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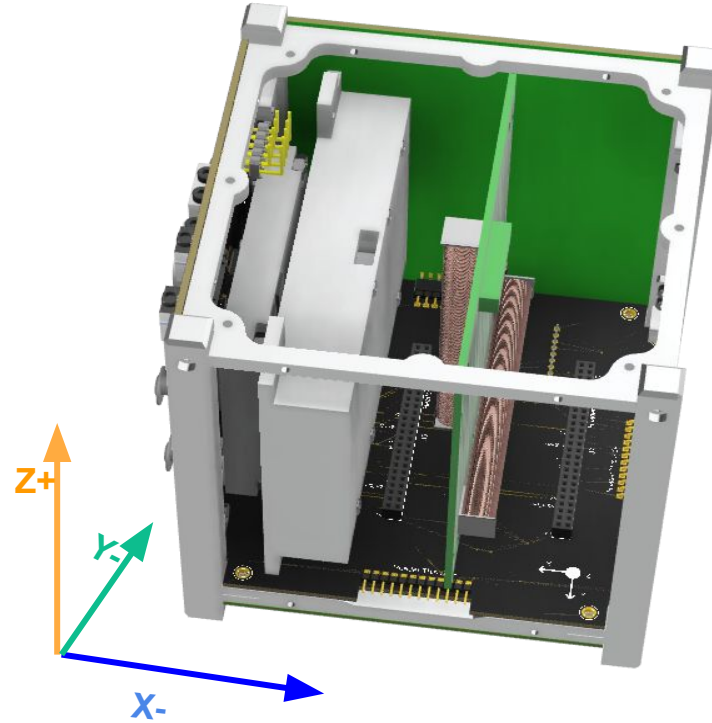
MSN3: Active Attitude Determination and Control System (ADCS)



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ADCS





Introduction



- ADCS (Active Attitude Determination and Control System)
 - Responsible for maintaining, monitoring and controlling satellite attitude
 - Includes the necessary sensors for the attitude determination
 - Involves use of actuators for controlling the satellite

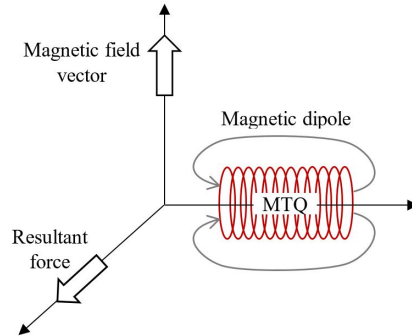


Illustration of Magnetic Dipole in Magnetorquer [\[2\]](#)



Magnetorquer [\[3\]](#)

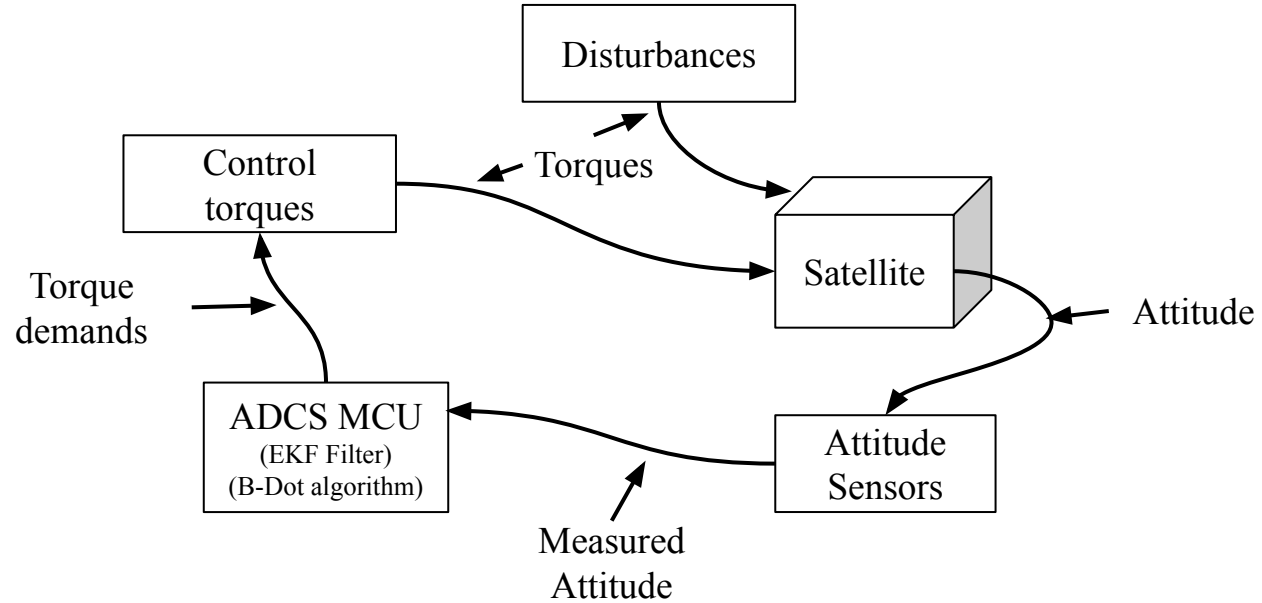


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Introduction



Block Diagram for ADCS [\[1\]](#)

EKF: Extended Kalman Filter
B-Dot: control algorithm



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Statement and Objectives



- Mission Statement

- To demonstrate the active satellite attitude control as the initial step for future APN satellite missions

- Mission Objectives

- Utilize attitude determination sensors to determine satellite orientation
- Employ magnetorquers to stabilize satellite
- Direct Satellite orientation towards Earth using magnetorquers for camera alignment



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Requirements (ADCS)



ID	System Requirements	Design Requirements	Verification Requirement
SR1.1	Satellite Attitude shall be determined by using Magnetometer, GPS, Solar vector, Gyro	Magnetometer shall be able to measure the magnetic field density in the range of 2500 nT - 6500 nT	Ensure datasheet states that the sensor can measure the magnetic field density in the range of 2500 nT - 6500 nT
		Shall use power generated from solar cells to predict the solar vector	Ensure to check the current and voltage of each solar cells
		3-axis Gyro sensor shall measure min. angular velocity of 0.1 deg/s	Ensure that sensor can measure min. of 0.1 deg/s angular velocity



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Requirements (ADCS)



ID	System Requirement	Design Requirement	Verification Requirement
		Shall receive location coordinates and time of Satellite from GPS	To make sure that satellite receive correct location coordinates and time from GPS
SR1.2	S2S shall perform detumbling in right direction to provide pointing accuracy for camera(± 5 degree)	Torque generated from magnetorquers shall interact with Earth's Magnetic field	North and South pole of the magnetorquer should be aligned properly
		Torque generated by magnetorquers shall be greater than disturbance torques from space environment($>4.58 \times 10^{-7}$ [Nm] i.e approx 2×10^{-6} [Nm])	Shall ensure the torque generated by magnetorquers is greater than disturbance torques on orbit

T =maximum moment by torque x maximum magnetic field



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Requirements (ADCS)

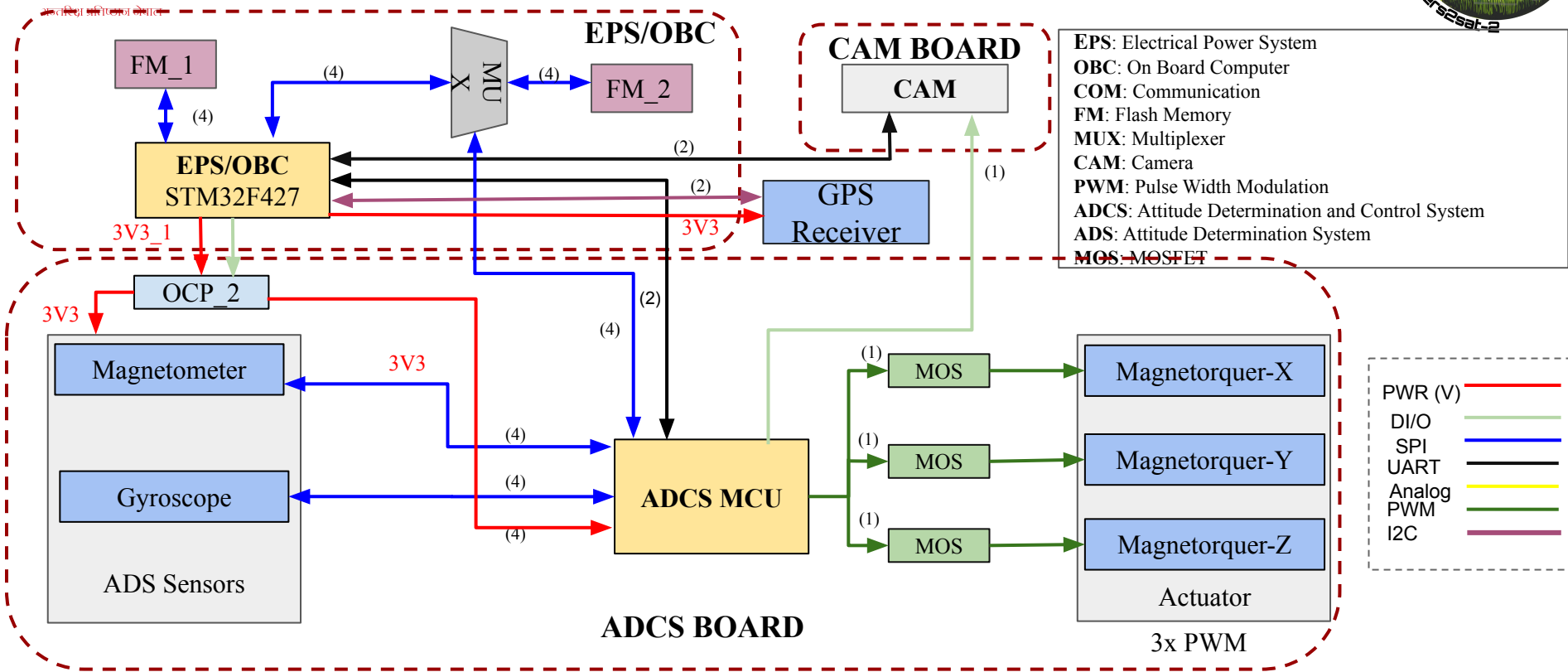


ID	System Requirement	Design Requirement	Verification Requirement
SR1.4	ADCS shall not affect other systems	Magnetic torque from magnetorquers shall not affect other systems	To make sure magnetic torque from magnetorquers shall not affect other systems
SR1.6	ADCS shall meet satellite limitations.	Size shall be included in area of 86mm x 90mm	Ensure measurements
		Mass of magnetorquers shall be less than 70 gm	Ensure mass
		The total power consumption shall be less than 0.7W	To make sure the power is enough for controlling attitude



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Block Diagram





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Components of ADCS



Hardware	Manufacturer	Main Features	Flight Heritage
Magnetometer (LSM9DS1)	STMicroelectronics	SPI/I2C interface, operational voltage 3.3V, magnetometer measurement range -200000 nT to +200000 nT	-
Gyroscope (MPU6500)	TDK	MEMS, operational voltage 3.3V, measurement range ± 250 deg/s	-
GPS Receiver (Venus838FLPx-L)	SKYTRAQ	Operational voltage 3.3V, data acquisition < 1 sec, low power	BIRDS4



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Components of ADCS



Hardware	Manufacturer	Main Features	Flight Heritage
Magnetorquers (NCTR-M003)	NewSpace	3-axis magnetorquer, operational voltage 5V , maximum magnetic dipole (0.29 Am²) (72mm x 15mm x 13mm), Power < 300mW	-
MCU STM32F446RE	STMicroelectronics	operational voltage 3.3V, PWM	BIRDS4
Sun Sensor (NXSS3V10)	needronix	FoV 110° x 110° , (15mm x 15.5mm x 4.6mm)	GRBALPHA



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Comparative Analysis of System Requirements



Components	Issues	Alternative Solution
Reaction Wheel (3 mNms RW-0.003)	33.5 x 33.5 x 17 mm , requires at least 3, 50g,	Magnetorquer
Sun Sensor (NXSS3V10)	9mm drill required between two solar cells (+Y, -Y, +Z or -Z axis) but 15mm x 15.5mm x 3.9 mm space requirement (Size and cost)	Measure solar vector using solar panel.



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Mission Modes



Mission Modes	Description	Energy Consumption Estimation
Attitude Determination Mode	Determine satellite orientation in space	1.161 Wh (0.036% duty cycle)
Detumbling Mode	Magnetorquers is used to actively control the orientation of satellite	1.163Wh (10.87% duty cycle)

No significant change in energy consumption as EPS/OBC is consuming high energy in both the modes.



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Feasibility Test (Size)



Mission Name	Components	Volume	Compatibility
Digipeater (MSN1)	NXT_GEN_CUBUS	50x100x113mm ³	YES
CAM (MSN2)	Imaging Sensor	6x6x0.75 mm ³	YES
	Lens x2	$[\pi \times (15)^2 \times (22.5) \text{mm}^3$ $(\pi r^2 h)] * 2$	YES
ADCS (MSN3)	Magnetorquer x2	72x15x14mm ³	YES
	GPS Receiver	25.14x25.13x4.4mm ³	YES
EPDM (MSN4)	Magnetometer x4	25.4x25.4x7.1mm ³	YES



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Feasibility Test (Weight)



Mission Name	Components	Mass	Compatibility
Digipeater (MSN1)	NXT_GEN_BUS	~0.476kgs	YES
CAM (MSN2)	Camera Sensor	0.017kgs	YES
	Lens	0.09kgs	YES
ADCS (MSN3)	Magnetorquer x2	0.06kgs	YES
	GPS Receiver	0.18kgs	YES
EPDM (MSN4)	Magnetometer x4	0.059kgs	YES

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Feasibility Test (Power)

**3SP E(30%) = 1173.6 mWh****3SP E(32%) = 1291.2 mWh**

BUS/ Mission	Telemetry			Mission			
	Burner	Beacon	HK-Data	DP-M	ADCS	CAM	EPDM
EPS/OBC	1136.66	1136.66	1136.66	1136.66	1136.66	1136.66	1136.66
COM TX (Beacon)	OFF	49.50430556	OFF	OFF	OFF	OFF	OFF
COM TX (Downlink)	OFF	OFF	129.1416667	OFF	OFF	OFF	OFF
COM RX (Uplink)	OFF	24.288	24.288	OFF	24.288	24.288	24.288
DP-M TX (Downlink)	OFF	OFF	OFF	129.1416667	OFF	OFF	OFF
DP-M RX (Uplink)	OFF	OFF	OFF	24.288	OFF	OFF	OFF
ADCS	OFF	OFF	OFF	OFF	1.739444444	1.739444444	OFF
CAM	OFF	OFF	OFF	OFF	OFF	25.10866667	OFF
EPDM	OFF	OFF	OFF	OFF	OFF	OFF	130.35
Burner Circuit	33	OFF	OFF	OFF	OFF	OFF	OFF
GPS	OFF	0.092	0.092	0.092	0.092	0.092	0.092
Total Energy (mWh)	1169.660	1210.452	1290.090	1290.090	1163.196	1187.796	1291.397



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Feasibility Test (Cost and Logistics)



Components	Cost	Logistic
Magnetometer (LSM9DS1)	\$5.88	Can be ordered from Mouser
Gyroscope (MPU6500)	\$8.71	Can be ordered from Mouser
Magnetorquers (NCTR-M003)	\$800-1200	USA office
Sun Sensor (NXSS3V10)	>\$1200	—
GPS Receiver (Venus838FLPx-L)	\$39.95	Can be ordered from DigiPart



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Success Criteria



ADCS Mission	Success Criteria
Minimum Success	Read attitude sensors data
Medium Success	Attitude stabilization is successful
Full Success	Satellite maneuver and orientation using magnetorquer